~\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\293C6C483966420D624234E5865437FD4D5B4047\transfers\2026-44\02_Exp2_U3.cpp

```
//                Name = Chandrashekhar Patil
//              RollN0 = 16
//                unit = 3
//    OPP Expriment No = 2









#include <iostream>
#include <string>
using namespace std;

// Multiple Inheritance
class PersonalDetails {
protected:
    string name;
    int age;
    string gender;

public:
    PersonalDetails() {
        name = "Unknown";
        age = 0;
        gender = "Unknown";
    }

    PersonalDetails(string n, int a, string g) {
        name = n;
        age = a;
        gender = g;
    }

    void displayPersonal() {
        cout << "Name: " << name << endl;
        cout << "Age: " << age << endl;
        cout << "Gender: " << gender << endl;
    }
};

class SalaryDetails {
protected:
    float basicSalary;
    float allowances;
    float deductions;

public:
    SalaryDetails() {
        basicSalary = 0;
        allowances = 0;
        deductions = 0;
    }

    SalaryDetails(float bs, float al, float de) {
        basicSalary = bs;
        allowances = al;
        deductions = de;
    }

    float calculateNetSalary() {
        return basicSalary + allowances - deductions;
    }

    void displaySalary() {
```

```
68          cout << "Basic Salary: " << basicSalary << endl;
69          cout << "Allowances: " << allowances << endl;
70          cout << "Deductions: " << deductions << endl;
71          cout << "Net Salary: " << calculateNetSalary() << endl;
72      }
73  };
74
75  class Employee : public PersonalDetails, public SalaryDetails {
76  private:
77      int employeeId;
78      string department;
79
80  public:
81      Employee() : PersonalDetails(), SalaryDetails() {
82          employeeId = 0;
83          department = "Unknown";
84      }
85
86      Employee(string n, int a, string g, float bs, float al, float de,
87               int id, string dept)
88          : PersonalDetails(n, a, g), SalaryDetails(bs, al, de) {
89          employeeId = id;
90          department = dept;
91      }
92
93      void displayEmployee() {
94          cout << "Employee ID: " << employeeId << endl;
95          cout << "Department: " << department << endl;
96          displayPersonal();
97          displaySalary();
98      }
99  };
100
101 // Hierarchical Inheritance
102 class Shape {
103 protected:
104     string color;
105
106 public:
107     Shape() {
108         color = "Unknown";
109     }
110
111     Shape(string c) {
112         color = c;
113     }
114
115     virtual void draw() {
116         cout << "Drawing a shape..." << endl;
117     }
118
119     virtual float area() = 0; // pure virtual function
120 };
121
122 class Circle : public Shape {
123 private:
124     float radius;
125
126 public:
127     Circle() : Shape() {
128         radius = 0;
129     }
130
131     Circle(string c, float r) : Shape(c) {
132         radius = r;
133     }
134
135     void draw() override {
136         cout << "Drawing a circle of radius " << radius << " with color " << color << endl;
137     }
```

```cpp

    float area() override {
        return 3.14159 * radius * radius;
    }
};

class Rectangle : public Shape {
private:
    float length;
    float width;

public:
    Rectangle() : Shape() {
        length = 0;
        width = 0;
    }

    Rectangle(string c, float l, float w) : Shape(c) {
        length = l;
        width = w;
    }

    void draw() override {
        cout << "Drawing a rectangle of " << length << "x" << width
             << " with color " << color << endl;
    }

    float area() override {
        return length * width;
    }
};

int main() {
    cout << "===== Multiple Inheritance - Employee =====" << endl;
    Employee emp("Alice Smith", 30, "Female", 60000, 15000, 5000, 1001, "IT");
    emp.displayEmployee();

    cout << "\n===== Hierarchical Inheritance - Shapes =====" << endl;
    Circle circle("Red", 5);
    Rectangle rectangle("Blue", 10, 6);

    circle.draw();
    cout << "Circle Area: " << circle.area() << endl;

    rectangle.draw();
    cout << "Rectangle Area: " << rectangle.area() << endl;

    return 0;
}
```